STQA PRACTICAL 6
SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project: Student Management System

1. Introduction

1.1 Purpose
This document defines the functional and non-functional requirements of a Student Management System. It serves as a reference for developers, testers, and users.

1.2 Scope
The web application maintains student records and allows authorized users to add, view, update, search, and delete student information.

1.3 Definitions
SRS: Software Requirements Specification; UI: User Interface; DB: Database.

2. Overall Description

2.1 Product Perspective: A standalone web application with a user interface, application logic, and database.

2.2 User Classes: Administrator manages records; staff may view or update records according to assigned permissions.

2.3 Operating Environment: A modern web browser on a computer or laptop; implementation may use HTML, CSS, JavaScript, PHP, and MySQL.

2.4 Assumptions: Users have browser and network access; the application server and database are available.

3. Functional Requirements

ID	Requirement
FR-01	The system shall allow authorized administrators to log in with valid credentials.
FR-02	The system shall allow the administrator to add a student record.
FR-03	The system shall display saved student records.
FR-04	The system shall allow the administrator to update student information.
FR-05	The system shall allow deletion of a selected record after confirmation.
FR-06	The system shall support searching by student name or ID.
FR-07	The system shall validate required fields and reject invalid input.
FR-08	The system shall allow authorized users to log out securely.

4. Non-Functional Requirements

ID	Requirement
NFR-01 Usability	The interface should be simple and easy to understand.
NFR-02 Performance	Common operations should respond within an acceptable time under normal load.
NFR-03 Security	Passwords must be stored securely and only authorized users may modify records.
NFR-04 Reliability	Valid changes should be saved correctly and clear errors shown when operations fail.
NFR-05 Compatibility	The application should work in current versions of common browsers.
NFR-06 Maintainability	The code should be modular and easy to update.

5. External Interface Requirements

User Interface: Login page, dashboard, student form, student list, and search field.

Software Interface: Web browser, application server, and database.

Hardware Interface: Standard computer or laptop capable of running a modern browser.

6. Data Requirements

Each student record may contain student ID, full name, date of birth, class or department, email address, and contact number. Student ID should be unique, and required fields must be validated.

7. Constraints

Only authorized users may modify records. The system depends on server and database availability and must follow institutional data-handling rules.

8. Acceptance Criteria

The system is accepted when authorized users can log in and out; add, view, update, search, and delete records; receive validation messages for invalid input; and save and retrieve records correctly.

9. Conclusion

This SRS documents the requirements of the Student Management System and provides a clear basis for design, development, and testing.